Rates and percents

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 548 to the nearest hundred.

2. Round 5.72 to the nearest whole number.

3. Use 20 × 5 to estimate 19.8 × 5.1.

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

5. A walker travels 12 miles in 3 hours. Find miles per hour.

6. 9 of 12 tiles are green. What percent are green?

7. 12 shells are 30% of a collection. Find the whole collection.

8. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

9. 2.4 + 1.75

10. 3.4 × 0.5

11. 4.4 + 1.75

12. 5.4 × 0.5

13. A walker travels 16 miles in 4 hours. Find miles per hour.

14. Omar says 24 is 40% of 9.6 because 0.40 × 24 = 9.6. Explain the error.

15. A cyclist travels 20 kilometers in 4 hours at steady speed. Find kilometers per hour.

16. 7 markers cost $3.5. Find dollars per marker.

17. A printer makes 18 cards in 3 minutes. Find cards per minute and minutes per card.

18. A runner goes 7 miles in 10 minutes at steady speed. Find miles per hour.

CHALLENGE / Explain your plan and check the result

C1. You have $15 and reserve 20% for a bus fare. Pens cost $1.50 each. What is the reserved amount, and what is the greatest number of pens you can buy with the rest?

C2. A tank contains 18 liters, which is 30% of its capacity. Water flows in at 6 liters per minute. How many liters are needed to fill it, and how long will that take?